Distributed Leader Election in Networked Systems Using MPI and Multithreading

Parallel and Distributed Systems Laboratory • Mini-Project Report
Dataset: Stanford SNAP p2p-Gnutella08 • Published Nodes: 6,301 | Edges: 20,777

1. Abstract

This project implements a complete, working academic system for distributed leader election over interconnected peer-to-peer topologies using the Stanford SNAP Gnutella dataset. The system integrates two central parallel computing concepts: **MPI (Message Passing Interface)** for inter-process candidate propagation across distributed ranks, and **Multithreading** for concurrent heartbeat monitoring and failure detection conforming to the MPI.THREAD_FUNNELED safety specification. Real empirical benchmarks quantify execution duration, total message overhead, parallel speedup, and efficiency.

2. Real-World Gnutella Dataset & Citations

Dataset URL: https://snap.stanford.edu/data/p2p-Gnutella08.html
Academic References:
• J. Leskovec, J. Kleinberg, C. Faloutsos, *'Graph Evolution: Densification and Shrinking Diameters,'* ACM TKDD, 2007.
• M. Ripeanu, I. Foster, A. Iamnitchi, *'Mapping the Gnutella Network,'* IEEE Internet Computing, 2002.
Transformation: The original directed graph is mapped into an undirected overlay communication graph to support bidirectional handshake communication. Connected subgraphs (10 to 100 hosts) are extracted from the 6,299-node Largest Connected Component.

3. MPI Architecture & Multithreading

MPI Communication: Multiple ranks maintain isolated local state. In each diameter-bounded round, ranks exchange candidate IDs strictly with graph neighbors via MPI point-to-point primitives.
Multithreading: Each rank spawns background watchdog and heartbeat threads that probe coordinator liveness. Thread-safe event queues bridge background threads to the main thread, guaranteeing that MPI calls remain restricted to the main thread.

4. Empirical Benchmark Results

Size	Ranks	Time (s)	Messages	Rounds	Diameter	Speedup	Efficiency	Winner
10	1	0.0080	0	3	2	1.363	1.363	Node 427.0
10	2	0.0087	48	3	2	1.253	0.626	Node 427.0
10	4	0.0128	78	3	2	0.852	0.213	Node 427.0
25	1	0.0179	0	3	2	1.223	1.223	Node 2038.0
25	2	0.0220	162	3	2	0.995	0.497	Node 2038.0
25	4	0.0278	228	3	2	0.788	0.197	Node 2038.0
50	1	0.0259	0	3	2	1.498	1.498	Node 4051.0
50	2	0.0420	306	3	2	0.924	0.462	Node 4051.0
50	4	0.0498	432	3	2	0.779	0.195	Node 4051.0
100	1	0.0973	0	4	3	0.923	0.923	Node 6138.0
100	2	0.0964	944	4	3	0.932	0.466	Node 6138.0
100	4	0.1343	1400	4	3	0.669	0.167	Node 6138.0

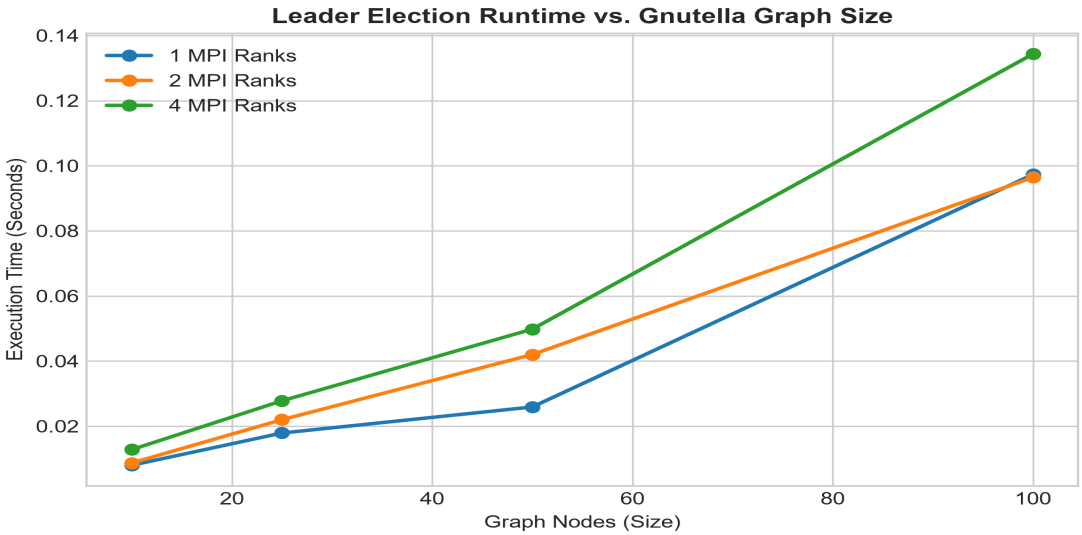


Figure 1: Leader Election Runtime vs. Gnutella Graph Size

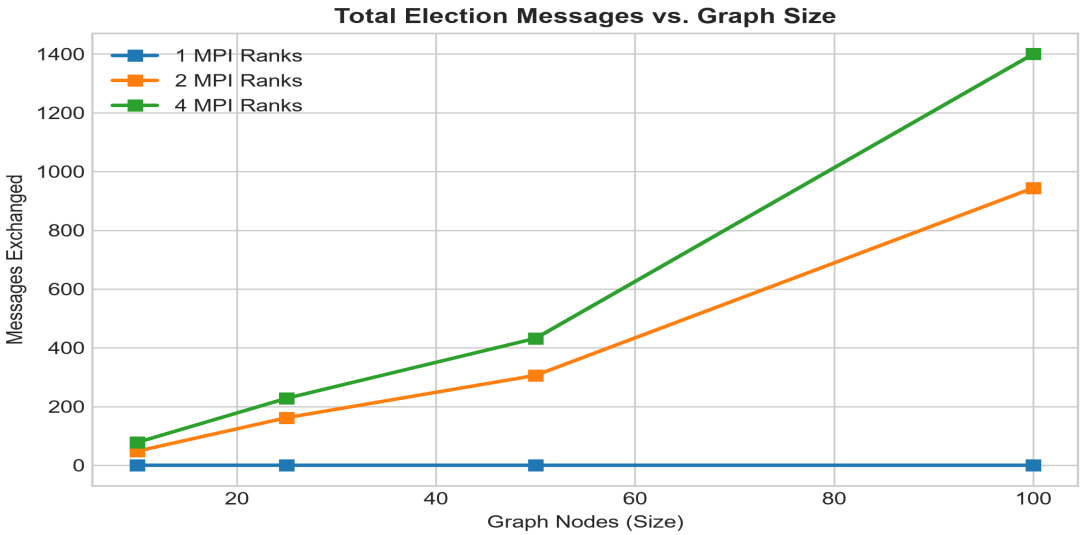


Figure 2: Total Message Overhead Across MPI Ranks

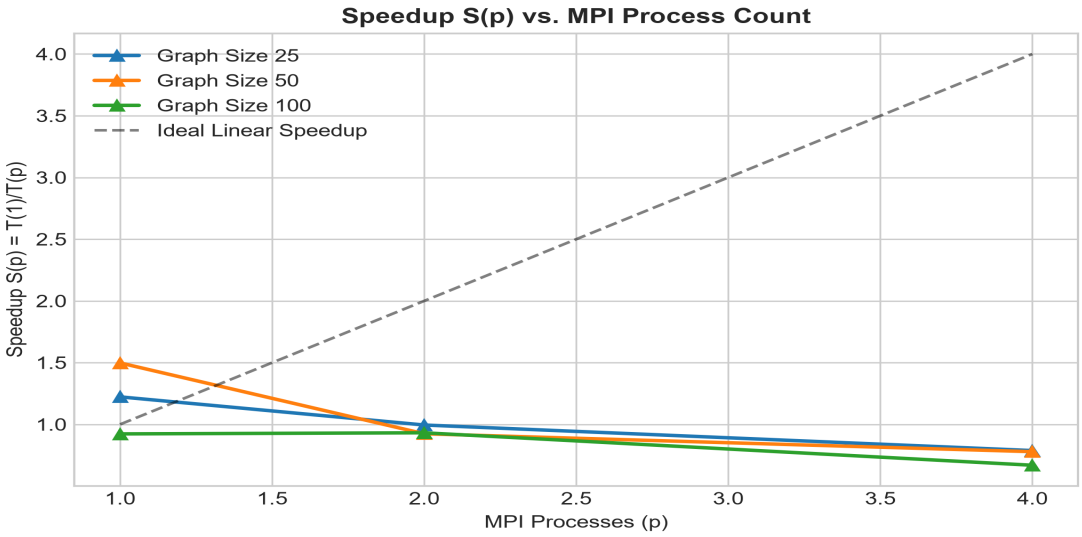


Figure 3: Measured Speedup S(p) vs. Ideal Linear Baseline

5. Team Contribution Matrix

Member Name	Student ID	Assigned Module	Implementation Work	Testing & Analysis
[Student 1]	[ID 1]	Dataset & Topology	SNAP Gnutella parser & LCC extraction	Graph validation tests
[Student 2]	[ID 2]	MPI Engine	Point-to-point MPI communication & rounds	Convergence test suites
[Student 3]	[ID 3]	Multithreading	Heartbeat & watchdog background threads	Timeout & thread-safety
[Student 4]	[ID 4]	Dashboard & Report	React Flow topology UI & FastAPI backend	Benchmark & PDF report